13. 3-Bet Pots and Balance

In this chapter we'll start by examining some crucial differences between 3-bet pots and the single raised pots we've been dealing with thus far. We'll then use the context of a 3-bet pot as a nice simplified realm where ranges are narrower to explore the real nature of balance. How do we apply it as the aggressor and as the defender and when should we deviate from it to maximise EV?

3-bet pots can be troublesome. They involve a stack to pot ratio we're less familiar with as 100BB deep cash players and any mistake that we make in them is magnified by the sheer size of every bet, call and raise going into the pot. We'll start out as the aggressor and talk about c-betting the flop. We'll discuss how sizing changes due to the difference in the effective stack and then move on to consider what some good ranges for value betting and bluffing and calling could look like Reg vs. Reg with the focus on balanced play.

13.1 C-Betting 3-Bet Pots

Implications of C-Bet Sizing

So far in the manual, we've been first considering whether or not to c-bet a hand on the flop and only then deciding upon our sizing. I'd like to reverse the process in this chapter. The effective stack size and ease with which we can get stacks into the middle in a 3-bet pot will have a large effect on the optimal c-bet size which will in turn have implications as to how wide we want to c-bet. Have a look at the following hand as an illustration.

Hand 126

CO – Reg (20/16/61/56)
BU – Unknown
SB – Hero
BB – Unknown

Dealt to Hero: X X

CO raises to 2.5BB, BU folds, **Hero raises to 8.5BB**, BB folds, **CO calls 6BB**.

Flop (18BB, 2 players) 7♣ 4♣ 4♦

Hero?

In a single raised pot, Hero would struggle to get stacks in by the river by making a standard two thirds pot sized bet. The pot might only be 6BB or so on the flop. Betting 4BB and getting called would leave a pot of 14BB on the turn and even a pot-sized bet there would leave a river pot of

42BB with an overbet 75-80BB left behind.

In a 3-bet pot there is no such issue and Hero's sizing, especially that of his flop c-bet, can be far smaller and still allow for the full amount to go in by the river. A bet of just 9BB here would create a pot of 36BB on the turn, which is very easy to work with in setting up a river shove.

In this hand, the hole cards are again marked with the sacred Xs that indicate that it's Hero's range and not any single part of it that will be the focus of our attention. Smaller c-bet sizing of around half pot is actually very agreeable to both Hero's light c-bets and his value c-bets. The former are always keen to decrease their RFE and make bluffing more profitable, while the latter are happy with any sizing that allows the money to go in over three streets. On wetter boards and vs. weaker opposition it can sometimes be better to build a bigger pot earlier with value hands since urgency of value is of higher importance and balance of lower importance. On a board this dry though, and against a Reg, all of Hero's c-bet range is comfortable betting 9BB.

Remember that a linear 3-betting strategy is usually advisable SB vs. BU since Hero doesn't want a flatting range by default unless he knows that BB rarely squeezes. This means that Hero will have an overpair much more often on this flop than Villain will and Hero's overcards will often dominate Villain's. This equates to significant range advantage for Hero. Villain will struggle to stop a half pot c-bet being +EV with any part of Hero's range. After all, it only has to succeed 33% of the time minus equity adjustment to break even (See Figure 23)

Hero bets 9BB.

It doesn't really matter what Hero has in this spot. A 9BB bet with his whole range is sensible and hard to exploit. This does not mean, however, that this strategy is always best. If Hero had reason to think Villain was unbalanced in some respect, then he could of course deviate from this plan. Against opponents who stab a lot of flops (have high **Float Flop** stats) Hero should develop a checking range and frequently check/call the flop. Against Villains who float (not the same as above) the flop relentlessly he could check/fold some hands and exploit them by betting a more value heavy range. The list goes on.

In summary, the main lesson here is that 3-bet pots are often more c-bettable than single raised pots as smaller sizing will always improve the RFE of light c-bets and there's no downside to this. There is no part of Hero's range that suffers from the pot growing more slowly.

C-Betting Light

We should already be experts on these decisions in single raised pots due to reading Chapter 4. It's now time to apply our knowledge to 3-bet pots. As we saw from the last hand, the following will be true.

The RFE on a Light C-Bet is usually lower in a 3-bet pot. Hero should assess the factors in the same way as in single raised pots, but be more lenient with what constitutes a good light c-bet for this reason.

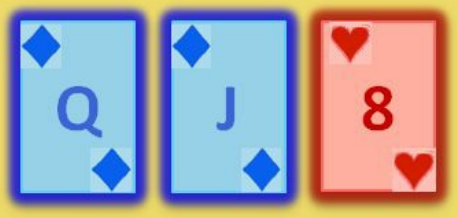
Lower RFE is a luxury afforded by pot size. Hero should capitalise by c-betting wider in most but not all cases. When then, might it be a bad idea to c-bet too wide in a 3-bet pot? Well, Villain's range to flat the 3-bet is better defined than his range to call an open would be. It's tighter and more uniform and so there are certain textures that it actually misses quite rarely. Take the following hand for example. The third (green) and fourth (purple) numbers on the HUD describe **Fold to PF 3Bet After Raise** and **Fold to F Cbet** respectively.

Hand 127

UTG – Unknown
HJ – Tight Reg (19/15/58/46)
CO – Hero
BU – Unknown
SB – Unknown
BB – Unknown

Dealt to Hero:

UTG folds, **HJ raises to 3BB**, **Hero raises to 8BB**, BU folds, SB folds, BB folds, **HJ calls 5BB**.

Flop (17.5BB, 2 players) 

HJ checks, Hero?

Let's start by giving this player a calling range to Hero's 3-bet from out of position. It will surely be

tight, capped, but still strong. His opening range shouldn't be wide to start with and we know that he folds a fairly large amount of the time to 3-bets hence Hero's polar approach including **A5s** as a bluffing hand.

AA	AKs	AQs	AJs	ATs	A9s	A8s	A7s	A6s	A5s	A4s	A3s	A2s
AKo	KK	KQs	KJs	KTs	K9s	K8s	K7s	K6s	K5s	K4s	K3s	K2s
AQo	KQo	QQ	QJs	QTs	Q9s	Q8s	Q7s	Q6s	Q5s	Q4s	Q3s	Q2s
AJo	KJo	QJo	JJ	JTs	J9s	J8s	J7s	J6s	J5s	J4s	J3s	J2s
ATo	KTo	QTo	JTo	TT	T9s	T8s	T7s	T6s	T5s	T4s	T3s	T2s
A9o	K9o	Q9o	J9o	T9o	99	98s	97s	96s	95s	94s	93s	92s
A8o	K8o	Q8o	J8o	T8o	98o	88	87s	86s	85s	84s	83s	82s
A7o	K7o	Q7o	J7o	T7o	97o	87o	77	76s	75s	74s	73s	72s
A6o	K6o	Q6o	J6o	T6o	96o	86o	76o	66	65s	64s	63s	62s
A5o	K5o	Q5o	J5o	T5o	95o	85o	75o	65o	55	54s	53s	52s
A4o	K4o	Q4o	J4o	T4o	94o	84o	74o	64o	54o	44	43s	42s
A3o	K3o	Q3o	J3o	T3o	93o	83o	73o	63o	53o	43o	33	32s
A2o	K2o	Q2o	J2o	T2o	92o	82o	72o	62o	52o	42o	32o	22

Call

Figure 76 - A Tight OOP Flatting Range vs. a 3-Bet

If Villain flats only these robust hands out of position here, which seems like a reasonable assertion, we can see that he absolutely nails flops like this one. The weakest hand he can have here is still a gutshot and two overcards (**AK**). The rest of his range is some kind of second pair or better and pair + draw or better and so Hero's immediate fold equity should be very low in Hand 127. It's not that Hero could never consider bluffing this flop, it's just that when he does he should probably be firing many turns and rivers as most of the fold equity he has vs. this range takes more than one street to realise. Such a situation caters for some selectivity of c-bet bluffs. It won't do to just take our whole

range to the river by tripling - this would result in barreling with terrible equity sometimes and being very unbalanced.

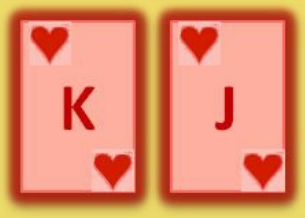
Hands with absolutely no equity when called like **As5s** then should simply be give-ups. Good RFE or not, Villain is just not folding enough of his range on this flop.

Hero checks.

This next spot is one where c-betting light is actually very good against Villain's range. What differences do you think make this the case?

Hand 128

UTG – Unknown
HJ – Unknown
CO – Unknown
BU – Reg (27/22/42/52)
SB – Unknown
BB – Hero

Dealt to Hero: 

UTG folds, HJ folds, CO folds, **BU raises to 2BB**, SB folds, **Hero raises to 7.5BB**, **BU calls 5.5BB**.

Flop (15.5BB, 2 players)

Hero?

The main difference here is that this board misses Villain's range really badly. A somewhat stationy Reg is going to be flatting very wide BU vs. BB and so will have an abundance of unpaired hands. Hero's deploys a wide linear value 3-bet range because he expects Villain to call too much.

One truth about NLHE is that it's much easier to be dealt a non-pair than a pair. For every pair there are only six combinations and the baby ones are rarely flatting the 3-bet due to how badly they perform in 3-bet pots. The top pairs are also unlikely guests in Villain's range as most players 4-bet as wide as **JJ+** in this late position battle spot. Every unpaired hand, however, is 16 combos and so these hands are much easier to hold.

As a result, Hero has such a significant range advantage that he can c-bet until his heart's content, even though he's out of position. Again, half-pot sizing makes for a nice low RFE of 33% before adjustment for realisation of equity and more fold equity in some branches when called. I'd reduce this number by a healthy 10% or so given our 6 good outs and backdoor flush draw. This would leave the true fold equity target at a tiny 23% or so. Hero's true RFE will be a little lower. This target should be incredibly achievable. To see why, let's visualise how Villain's range connects on this texture.

AA	AKs	AQs	AJs	ATs	A9s	A8s	A7s	A6s	A5s	A4s	A3s	A2s
AKo	KK	KQs	KJs	KTs	K9s	K8s	K7s	K6s	K5s	K4s	K3s	K2s
AQo	KQo	QQ	QJs	QTs	Q9s	Q8s	Q7s	Q6s	Q5s	Q4s	Q3s	Q2s
AJo	KJo	QJo	JJ	JTs	J9s	J8s	J7s	J6s	J5s	J4s	J3s	J2s
ATo	KTo	QTo	JTo	TT	T9s	T8s	T7s	T6s	T5s	T4s	T3s	T2s
A9o	K9o	Q9o	J9o	T9o	99	98s	97s	96s	95s	94s	93s	92s
A8o	K8o	Q8o	J8o	T8o	98o	88	87s	86s	85s	84s	83s	82s
A7o	K7o	Q7o	J7o	T7o	97o	87o	77	76s	75s	74s	73s	72s
A6o	K6o	Q6o	J6o	T6o	96o	86o	76o	66	65s	64s	63s	62s
A5o	K5o	Q5o	J5o	T5o	95o	85o	75o	65o	55	54s	53s	52s
A4o	K4o	Q4o	J4o	T4o	94o	84o	74o	64o	54o	44	43s	42s
A3o	K3o	Q3o	J3o	T3o	93o	83o	73o	63o	53o	43o	33	32s
A2o	K2o	Q2o	J2o	T2o	92o	82o	72o	62o	52o	42o	32o	22

Folding Very Often

Folding Often

Folding Rarely

Folding Never

Figure 77 - A Wide BU Flatting Range vs. SB 3-Bet

This range is fairly standard for a loose Reg to defend with in position on the BU, if not even a little conservative. We can see from the distribution of his range on this flop how rarely Villain has any kind of one pair or better. The darker green group has nothing but two raw overcards and is not advised to continue without reads that we are c-betting then giving up too much. The lighter green

group has usually flopped a backdoor flush draw and overcards to the board and so will float sometimes. Villain must be careful though as these overcards are often dominated and hitting them can be as much of a curse as it is a blessing vs. our uncapped linear range. We still expect to get a good amount of folds from this group. The yellow group has sufficient SDV to usually want to peel the flop but it's not unheard of for Regs to fold these hands. Finally, the red group is one pair plus and pretty much never going anywhere.

The eager combo counter should take this moment to count up what percentage of Villain's range each of these groups constitutes. The result will show how much more fold equity we have than the measly target we have to surpass when we bet half pot usually with two overcards or better when called. The average Villain with these stats will certainly fold much more than 23% here. While this doesn't automatically make c-betting the highest EV line in and of itself, I can't imagine checking behind being better. C-betting everything is a good way to utilise range advantage, make lots of high EV c-bets in a vacuum and play an unexploitable strategy.

Hero bets 8BB

13.2 3-Bet Pots As The Aggressor

Value-Betting 3-Bet Pots

Just as we saw in single raised pots in Chapter 12, if starting out balanced is our concern, then we'll need to choose the most appropriate segments of our range to bluff/semi-bluff with as well as deciding which hands we can value bet on each street. One central concept here is that value ranges for getting stacks in should be wider in 3-bet pots than they would be in single raised pots due to the smaller effective stack making players more committed. As we just saw in a 3-bet pot, 100BB stacks end up in the middle just by one player betting three times and so getting all in is very achievable without getting raised.

Let's start with an example of a bread and butter situation where barreling off for value is standard.

Hand 129

UTG – Unknown
HJ – Unknown
CO – Reg (24/19/55/56)
BU – Hero
SB – Unknown
BB – Unknown

Dealt to Hero:

UTG folds, HJ folds, **CO raises to 2.5BB**, **Hero raises to 7.5BB**, SB folds, BB folds, **CO calls 5BB**.

Flop (16.5BB, 2 players)

CO checks, **Hero bets 8BB**, **CO calls 8BB**.

Turn (32.5BB, 2 players)

CO checks, **Hero bets 19BB**.

River (70.5BB, 2 players)

CO checks, **Hero bets 66.5BB (all in)**.

The main advantage Hero wields here as the pre-flop 3-bettor is that of having an uncapped range

against a capped range. It is very likely that Villain 4-bets **AK** pre-flop and so these combos should be heavily discounted. If **AK** is so unlikely, then what better hand does Villain get to the river with? Given his moderate 55% **Fold to 3Bet After Raise** stat, it's also doubtful that he calls the 3-bet with **A6s** or **A8s**, though the latter in particular is certainly far from impossible. If Villain has any better hands at all in his range than **AQ**, then they are few and far between. This grants Hero free reign to value shove this river against all of the more abundant **AJ**/**AT** combos without fear of value owning himself.

The more often that Hero has the best hand, the thinner he can afford to value bet since even if Villain calls only rarely with a worse hand, there are so few hands that Hero loses to.

3-bet pots can justify stacking off with weaker absolute strength hands for two reasons:

- The relative stack size is smaller and so Villain should be inclined to stack off with a weaker range himself.
- Ranges for calling 3-bets are better defined than ranges for calling opens and as a result are frequently capped to exclusively worse hands. This allows Hero to value bet with impunity even with just top pair second best kicker.

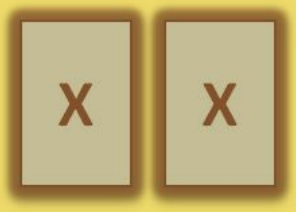
Bluffing in 3-Bet Pots

Constructing bluffing ranges can be a tricky task in a single raised pot due to the sheer amount of combos in Hero's opening and c-betting ranges. In a 3-bet pot this task becomes a little simpler. Ranges are smaller and better defined and so are easier to work with whether striving for balance or some exploitative ratio of bluffs to value.

3-bet pots serve as a nice simplified environment in which to get our feet wet with this type of exercise. Let's take a look at how Hero might select his bluffs in a 3-bet pot where he is very aware of his own 3-betting range.

Hand 130

UTG – Reg (21/17/68/50) (655 hands)
HJ – Unknown
CO – Unknown
BU – Unknown
SB – Unknown
BB – Hero

Dealt to Hero: 

UTG raises to 3BB, HJ folds, CO folds, BU folds, SB folds, **Hero raises to 10BB**, **UTG calls 7BB**.

Flop (20.5BB, 2 players)

Hero bets 9BB, **UTG calls 9BB**.

Turn (38.5BB, 2 players)

Hero bets 22BB, **UTG calls 22BB**.

River (82.5BB, 2 players)

Hero bets 59BB (all in).

Okay, so here's a step by step template of how we decide what hands to bluff all-in with on the river.

Q1. What is Hero's pre-flop 3-betting range?

Q2. How many combos are we barreling off with for value on this river?

Q3. Given this, how many bluff combos would give us a balanced ratio of bluffs to value.

Q4. What should these bluffs be?

Q1. What's our pre-flop 3-betting range?

Hero should clearly have a polarised strategy here and one that is more bluff heavy than balance would call for due to Villain's tendency to overfold vs. 3-bets. As we saw in Chapter 10 Hand 87, he should strive to 3-bet a tight value range vs. such a player opening from UTG and use the best hands he'd otherwise be folding as bluffs as per the polar model.

His range could look something like this:

AA	AKs	AQs	AJs	ATs	A9s	A8s	A7s	A6s	A5s	A4s	A3s	A2s
AKo	KK	KQs	KJs	KTs	K9s	K8s	K7s	K6s	K5s	K4s	K3s	K2s
AQo	KQo	QQ	QJs	QTs	Q9s	Q8s	Q7s	Q6s	Q5s	Q4s	Q3s	Q2s
AJo	KJo	QJo	JJ	JTs	J9s	J8s	J7s	J6s	J5s	J4s	J3s	J2s
ATo	KTo	QTo	JTo	TT	T9s	T8s	T7s	T6s	T5s	T4s	T3s	T2s
A9o	K9o	Q9o	J9o	T9o	99	98s	97s	96s	95s	94s	93s	92s
A8o	K8o	Q8o	J8o	T8o	98o	88	87s	86s	85s	84s	83s	82s
A7o	K7o	Q7o	J7o	T7o	97o	87o	77	76s	75s	74s	73s	72s
A6o	K6o	Q6o	J6o	T6o	96o	86o	76o	66	65s	64s	63s	62s
A5o	K5o	Q5o	J5o	T5o	95o	85o	75o	65o	55	54s	53s	52s
A4o	K4o	Q4o	J4o	T4o	94o	84o	74o	64o	54o	44	43s	42s
A3o	K3o	Q3o	J3o	T3o	93o	83o	73o	63o	53o	43o	33	32s
A2o	K2o	Q2o	J2o	T2o	92o	82o	72o	62o	52o	42o	32o	22

3-Bet Value 3-Bet Bluff

Figure 78 - A Bluff Heavy Polar 3-bet Range BB vs. UTG

Hero's bluff portion of his range outweighs his value portion by 40 combos to 12 or 3.33:1. No doubt this range is very exploitative, but 68% is a huge fold to 3-bet stat and so such an imbalance is easily justifiable. Bluffs are once again chosen from the best of the 'would be fold' hands based on which have the best blockers + post-flop playability. **[A2s-A5s]** here should outperform **[A8s-A9s]** due to their extra playability and similar pair flopping power. Villain is unlikely to be defending weak **Ax** or medium pocket pairs like **77** and **88** here and so whether or not we have a **9** or a **5** with our ace does not matter from a pair-making perspective. It does matter from a straight making perspective though and only the wheel aces have a shot at doing this.

Q2. What's our river value range?

Here our task is very simple indeed, we have 12 combos of overpairs and then 4 combos of straights (**A3s**) and these are the only value bettable hands in our range on this river. That gives us a total of 16 value combos. Let's now work backwards and see what hands we would like to bluff in order to create a balanced strategy.

Q3. How many bluffs would give us a balanced river betting range?

To answer this question we first need to acquaint ourselves with what we're going to be calling the indifference principle.

The Indifference Principle

1. Whenever Villain will be 0EV by calling a bet with the middle of his range, Hero's betting/raising range is balanced.

2. Whenever Villain will be 0EV by bluffing, Hero's calling range is balanced.

We implicitly met this concept already in Chapter 11 when we designed balanced defense ranges. The indifference here is that Villain does not profit from choosing one option over another and this means that he cannot exploit Hero through making his choice. If calling and not calling are equal in EV in Hand 130, then Hero has achieved a balanced betting range i.e a range that cannot be exploited.

Such a strategy is defensive in nature and should be adopted only where Hero does not have a higher EV exploitative strategy at his disposal. The thought is that by becoming balanced, Hero gives away no opportunity to be exploited, and since he himself has no route to exploiting Villain, this is the best he can do. Why give Villain imbalances to attack if you're not gaining anything in the process? It's only when we figure to gain by being unbalanced that we can justify granting such an opportunity.

Back to our model hand: why do we want a balanced strategy here? Because we have no relevant information on Villain with which to determine a way that he is unbalanced post-flop. Since we do not known whether Villain is overcalling or undercalling his range on the river, we don't have a reason to stray from balance. Beware of the following fallacy.

The **Readless Therefore Nit Fallacy (RTN Fallacy)** is the belief that Hero should refrain from bluffing until he knows more about his opponent. All this does is create a very unbalanced range that underbluffs, offering Villain a route to exploit Hero by overfolding. There is no reason to do this unless the population grossly underfolds its range in the spot in question.

Some players are terrified of bluffing incase it turns out that they had no fold equity, but it could just as easily turn out that there was lots of fold equity and that not bluffing was a big mistake. Don't be unbalanced for absolutely no reason other than irrational fear - this is rightly forgotten 2004 thinking.

How then do we create balance with respect to our bluff combos? We follow the balancing principle below.

Balancing Principle 1

Villain is indifferent to calling Hero's bet in an end of action spot with a bluff catcher whenever Hero's bet size generates an RE target of X.

AND

Hero's ratio of bluff to value combos in a polarised betting range gives Villain exactly X equity with a bluff catcher.

Recall that RE describes the equity % of Villain's hand vs. Hero's range that will cause Villain to break even on a call in an end of action spot. It is his bare minimum equity for calling a bet from Hero.

In Hand 130 Hero's bet-size is a shove as the effective stack is small enough for shoving to be feasible and there is no reason to suppose any other bet-size will be better with Hero's range. Generally speaking, polarised ranges on the river want to bet big as this will generate a higher RE target and thus allow for more bluffs in Hero's range.

Hero bets 59BB into a pot of 82.5BB. What is Villain's RE?

RE = ATC / (ATC + TP)

RE = 59 / (59 + 141.5)

RE = 29.4%

Following the balancing principle, this means that 29.45% of Hero's river betting range should be a bluff. Since his betting range contains 16 value combos we can solve for his required bluffs (X).

X / (16 + X) = 0.294 **(1)**

X = 0.294 (16 + X) **(2)**

X = 0.294X + 4.704 **(3)**

(X - 0.294X) = 4.704

0.706X = 4.704

X = 4.704 / 0.706

X = 6.662

(1) We know that X will be 29.4% of Hero's total betting range. Hero's total betting range will be X + his value combos (16 + X).

(2) Simplification: multiply both sides of the equation by (16 + X).

(3) Simplification: multiply both 16 and X by 0.294 to remove brackets.

For the algebraically shy, there are readily available algebra calculators online for such a job. Let's round this fractional answer up to the nearest whole number and conclude that Hero needs **7 bluff combos** to create the 7:16 or 0.44:1 ratio of bluffs to value that satisfies the balancing principle.

Q4. What should these bluff combos be?

This firstly depends on what bluff combos Hero gets to the river with. Let's say that he continues bluffing on the turn with **[A2s-A3s]** (8) and one third of the combos of **KQo** (4). Hero value bets **KK** and **AA** (12) and so has a turn range of **12 value combos and 12 bluffs**.

So we need just 7 combos of bluffs and we have the choice between either **KQo** or **A2s**, which has now made a pair. How are we to select between these combos? There are two central guidelines when it comes to choosing bluffs to flesh out our range.

1. Hands with less SDV are better to bluff than hands with more SDV.

And

2. Hands that block Villain's calling range and not his folding range are better bluffs than the opposite.

The first point is simple and is to do with the EV gain of bluffing. See the box below for a detailed explanation, but feel free to skip it for now.

EV Gain of Bluffing

We always compare the EV of a bluff to the EV of the more passive alternative (checking or calling) and not to 0EV. This tells us how much better bluffing is than not bluffing.

Checking and going to showdown some % of the time with SDV, even with the intention of folding to a bet, is always better than 0EV as Hero wins some amount of money some % of the time from the pot.

If Hero lost all of the pot all of the time (had no SDV) then checking would be 0EV from point of decision.

When Hero checks a hand with no SDV his EV is 0.

Bluffing that hand needs to be better than 0EV to be the best line i.e be successful RFE% of the time or more.

When Hero checks a hand with some SDV his EV is +X depending on how often he wins the pot and the pot size.

Bluffing that hand needs to be better than +X EV to be the best line and therefore be successful some amount more than RFE% of the time to be the best line.

This is why we favour lower SDV hands when we bluff.

The second point is to do with card removal. If Hero blocks the combos Villain would have called with, then Villain folds a higher % of his range and Hero's bluff is higher EV. If Hero blocks the combos Villain would have folded with, then Villain calls a higher % of his range and Hero's fold equity is less and his bluff lower EV.

So let's apply this criteria to our river decision in Hand 130. Which combos of the bluff candidates in our range best satisfy these requirements? Recall that we bet the turn with **KK**(6), **AA**(6), **A2s**(4), **A3s**(4) and **KQo**(4), but on the **2h** river, **A2s** has become only 3 combos and **A3s** has become the nuts and goes into our value range, as we saw.

The only possible bluffs we now have are our 4 **KQo** combos and our remaining 3 **A2s** combos. Conveniently, this meets our 7 combo bluff target exactly! To be balanced here, Hero needs to follow through with every bluff in his range. The fact that he shouldn't give up with any bluffs would normally make him unbalanced on most river cards, but not ones like this, which help his turn bluffing range tremendously. Let's not just stop our analysis there though, which hand would be the better choice if Hero needed less than 7 bluffs and did want to give up some combos?

Well it's fair to say that neither **KQo** nor **A2s** has much SDV vs. a range that Villain calls flop and turn with, but maybe **A2s** wins vs. Villain's **AK** sometimes. This alone makes it worse to bluff.

Blocker wise, **KQo** is also clearly better. This hand blocks **KK** and **QQ**, two hands that Villain is fairly likely to call the river with. Allowing likely folding hands like **TT** and **99** to be a bigger part of Villain's range. Hero would prefer **KQo** to **A2s** here, but as is he'll be betting both.

Conclusion

This exercise has been highly theoretical and is not something I recommend trying in game. Doing work like this in 3-bet pots is a nice place to get started with balancing complete post-flop ranges since 3-bet pot ranges are snugger and easier to work with. Rinse and repeat this process out of game for a multitude of different 3-bet pot scenarios and your in-game feel will gradually improve allowing you to play closer to balanced on the fly.

Balancing Betting Ranges Before the River

Before we move on, let's return to our turn range in Hand 130. This is not an end of action spot so the correct balanced strategy is not so easy to solve for. This is because Hero's bluffs are not 0 equity hands vs villain's bluff catchers, they can and often do improve! How does this fact change things when we solve for villain's RE and use it to satisfy the indifference principle? See the semi-bluffing principle below:

The Semi-Bluffing Principle

Hero should add more bluffs to his balanced betting range on the flop or turn when:

1. His bluffs have more outs vs. Villain's bluff catchers.
2. There are more streets to come.

This is because when Villain is considering calling a bet on earlier streets, he needs to consider that Hero's bluffs have equity vs his bluffcatchers and so Villain's RE target increases. Villain does not have either 0% or 100% equity with a bluff catcher with cards still to come as he would on the river. He is crushed by Hero's value hands and Hero's bluffs still have some equity. He might instead have something like either 8% or 80% equity depending on whether Hero is bluffing.

As an approximate guide, Hero should adjust Villain's RE upwards by around **20% on the flop** and **10% on the turn**, but the exact amount will depend on how many outs Hero's bluffs have.

This allows Hero to approximate balance in open action spots.

So on the turn in Hand 130 we were betting 22BB into 38.5BB. Villain's RE before adjustment would be 22 / (22 + 60.5) = 27%.

But, we must adjust this to account for Hero's equity with his **KQo** and **[A2-A3s]**. The former has 6 outs vs. bluff catcher hands like **JJ** while the latter has 7 outs. As per the rule above, we'll add 10% onto this target and give villain a 37% equity target to break even on calling turns.

Recall that Hero ended up betting 1:1 bluffs to value hands meaning he was bluffing 50% of the time. This makes calling the turn with a bluff catcher +EV for villain. If Hero wanted to still bet this range and make Villain more indifferent to calling then he'd have to increase his sizing a bit to hurt Villain's RE. We can't always be perfectly balanced. Note that if Hero was to subtract some **KQo** from his turn bluffing range to achieve perfect balance on the turn, he'd no longer be able to bluff enough on the 2h river.

Sometimes board run outs can just unbalance ranges and it's hard to get your sizing and range construction exactly right in-game. As long as we try our best to emulate optimal balanced strategies where desirable, we'll be on the right track. The goal is to be aware of our range and use it along with our sizing to come as close as we can to balance where that is a desirable end. A lot more money is made from good exploitative play than getting balance right down to the decimal point!

In-Game Balance Examples

Let's now look at a few examples where we have exact hole cards before moving on. This will help us approximate the above kind of thought process in-game. In the following hands, try to decide if Hero should bluff the street in question or if there is a better alternative.

Hand 131

UTG – Unknown
HJ – Unknown
CO – Unknown
BU – Likely Reg (4 hands)
SB – Hero
BB – Unknown

Dealt to Hero:

UTG folds, HJ folds, CO folds, **BU raises to 2BB**, **Hero raises to 7.5BB**, BB folds, **BU calls 5.5BB**.

Flop (16BB, 2 players)

Hero bets 8.5BB, BU calls 8.5BB.

Turn (33BB, 2 players)

Hero?

A likely Reg is a player who has done something over a very small sample that Regs will commonly

do, but Fish won't. This will usually be some sizing related clue, perhaps Villain 2.5x opened his SB a couple of hands ago.

Hero 3-bets a linear but fairly wide range here SB vs. BU. The absence of any kind of Fish or weaker player in the BB would deter him from 3-betting polar even if he suspected that the Reg population folds to too many 3-bets. Hero simply doesn't have a flatting range by default here.

On the flop then, Hero has clear range advantage since his range contains **[JJ+]** where as Villain's capped range most likely does not. Hero can also hold **AK** where as Villain usually can't. **AK** is a surprisingly strong hand on this texture since it dominates Villain's air, has 6 outs vs. all of Villain's pocket pairs and has a very substantial degree of SDV. Many of Hero's one pair hands are in need of protection here on this texture and so Hero should simply bet his whole range on this flop. Villain will not be equipped to deal with it given his range disadvantage. This way Hero can delay forming a checking range until later in the hand and make sure he protects his equity advantage.

So Hero reaches the turn with his whole pre-flop range. **AsKs** is a rare part of that range that has both SDV and a nutted draw. How should it be played? Well, since we know of no exploitative route to take here we should be aiming at having a balanced strategy and bluffing/semi bluffing the turn with slightly less combos than we'd value bet. Recall from the Semi-Bluffing Principle that on the turn we may use a higher concentration of bluffs than on the river.

We will also need a check/call range as well as a betting range as undoubtedly there are now some hands with poor equity that we'd rather give up with and so we now have a check/folding range that needs balancing. We must not fold the turn a disproportionate amount of the time when we check. Checking hands is now far more acceptable than it would have been on the flop as Villain's flop call narrowed his range meaning we no longer have such an advantage range vs. range by the time we get to the turn.

What hands should go into the check/call range then? Certainly, SDV is our first and foremost requirement and **AK** has plenty of this vs. Villain's flop floats and **A**-high SDV calls. In fact, it dominates a great portion of those hands. While betting the flop with **AK** was probably technically a value bet for Hero, betting the turn would not be; it would be to turn a high SDV hand into a bluff, something that a balanced strategy will usually have us avoid. Normally **AK** would be a clear check/call on this turn.

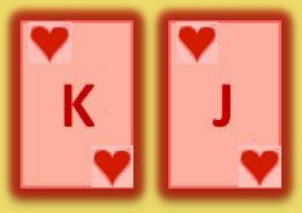
Does our flush draw change things though? No, not at all, in fact it helps the check/call plan further. Villain is probably not folding many better hands than this one even if we bet turn and river and so semi bluffing is just not enticing with this part of our range. Having a flush draw is actually quite nice in the sense that it puts some flushes in our c/c turn range those times a spade falls on the river. Hero should check the turn and look to gladly call a bet. He can bluff weaker hands than this and value bet stronger ones. What better hand to balance his check/folding range with?

Sure it's +EV to bet, but it's part of the highest EV long-term strategy to check.

Hero checks.

Hand 132

UTG – Unknown
HJ – Unknown
CO – Unknown
BU – Unknown
SB – Aggro Reg (31/24/35/36) (4k hands)
BB – Hero

Dealt to Hero: 

UTG folds, HJ folds, CO folds, BU folds, **SB raises to 3BB**, **Hero raises to 9BB**, **SB calls 6BB**.

Flop (18BB, 2 players)

Hero bets 9BB, **UTG calls 9BB**.

Turn (36BB, 2 players)

Hero bets 21BB, **UTG calls 21BB**.

River (78BB, 2 players)

Hero? [70BB left]

This hand could well be a call pre-flop against most opponents. Few Regs are flatting as wide as this

guy is out of position here though and so KhJh can serve as a value 3-bet as part of a linear range.

The flop is an unbelievably easy c-bet. If we're not bluffing this hand with it's frequent two overcards to many pairs, backdoor flush draw and gutshot, then we are simply not bluffing the flop anywhere near enough in position. SDV is insufficient to check. On the turn the same is true. We now pick up so much equity that our hand is now possibly the most suitable turn bluff in our entire range if we don't 3-bet **JhTh**. This street is also a must bet for similar reasons unless Hero wants to check his whole range and he really doesn't given that he has a lot of **Qx** that wants to value bet three times against pocket pairs and **9x**.

Hero's river shove would be for almost pot, which we now know mean that he needs a ratio of just over 2:1 value to bluffs to be balanced since these are the odds he's offering to Villain by shoving. Hero's value range is likely no weaker than **Qx** for value (anything else would be too thin). If Hero 3-bets only **KQ** and **AQ** pre-flop (**QJs** should be okay too though), then he has 16 combos of value. This means he needs around 7 bluff combos and so should select only the most suitable hands for the job.

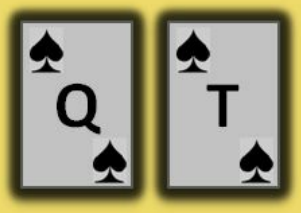
This hand blocks **KQ** and **QJ**, which are undoubtedly a sizeable part of Villain's river calling range. This is a first class bluffing candidate and so should be bluffed even though Hero's value range is small.

The only reason not to bluff here would be if we thought Villain was too stationy to fold even the lower parts of his range. There could be an argument for this given these stats, but Hero should have built up some player notes over the 4k sample and should now refer to these to see if such an extreme exploitative adjustment is warranted. Otherwise...

Hero shoves.

Hand 133

UTG – Unknown
HJ – Unknown
CO – Unknown
BU – Fish (44/24/20/58) (800 hands)
SB – Hero
BB – Unknown

Dealt to Hero: 

UTG folds, HJ folds, CO folds, **BU raises to 3.5BB**, **Hero raises to 10.5BB**, BB folds, **BU calls 7BB.**

Flop (22BB, 2 players)

Hero bets 9.5BB, BU calls 9.5BB.

Turn (41BB, 2 players)

Hero?

Pre-flop is a clear 3-bet yet many of my students might make the mistake of flatting this hand. They may argue that they should 3-bet an exclusively value 3-bet range and that **QTs** is not good enough to be a part of this range. This thinking is mistaken here due to one saving factor - post flop fold equity.

It is far better to inflate the pot and isolate a bad player with a very playable hand while gaining the initiative to harvest post-flop fold equity than it is to flat and potentially get squeezed out of the pot and miss some pre-flop fold equity too. Hero will also win the pot much less often when he sees the flop as the pre-flop caller. Hero must 3-bet here!

The flop c-bet is good given all that pot equity and definitely a fair bit of fold equity too. Some people are terrified of this flop. They might protest:

"But this flop is all over his range."

This is just the Fish Flop Fallacy in action again.

This turn is a pretty clear bet in any spot where Hero craves balance - he just doesn't have many better bluffs in his range. Balance is one of the last things we care about here though. This player is very unlikely to be folding enough of his flop calling range on this anonymous turn card for betting to be better than checking. By checking we accept that betting is not going to be much better than 0EV, and stop ourselves from ever getting blown off our equity, which is fairly disastrous. Checking is clearly a good bit better than 0EV due to all the times we get there on the river and win the pot as it stands or, better yet, a further bet. Don't bluff Fish on non-board changing turns; this is one very reliable rule if there is such a thing in this game.

Hero checks.

13.3 Strategy As The Defender

As the defender post-flop we have a similar route available to achieving balance and satisfying the indifference principle that we met earlier. We are now working with the second part if it: **"Whenever Villain will be 0EV by bluffing, Hero's calling range is balanced"**, that is to say that Hero is neither undercalling nor overcalling his range.

Calling hands should be chosen again using two central criteria.

1. Hero should elect to call the hands with the most SDV.

And often more importantly:

2. Hero should look to call the hands with the best blockers to Villain's value betting range and least blockers to his bluffing range.

It's a symmetrical situation to bluffing. Hero wants to bluff catch with the hands that make bluffs the most likely type of hand for Villain to hold and that involves blocking the non-bluffs. There are many common situations where Villain's betting or raising range is totally polarised. This means that each bluff catcher combo in Hero's range has exactly the same SDV as the others. If Villain can only have a flush or air, then a straight is as good as a pair, it's the one that blocks flushes the most heavily that takes priority in the calling range.

Let's look at an example of how to defend our range facing a bet on the river as this is the simplest street to assess in terms of how to play a balanced range.

Hand 134

UTG – Unknown
HJ – Unknown
CO – Hero
BU – Unknown
SB – Unknown
BB – Unknown (0 hands)

Dealt to Hero: X X

UTG folds, HJ folds, **Hero raises to 2.5BB**, BU folds, SB folds, **BB raises to 8.5BB**, **Hero calls 6BB**.

Flop (17.5BB, 2 players) A♣ 8♥ 8♦

BB bets 10BB, **Hero calls 10BB**.

Turn (37.5BB, 2 players) 7♠

BB bets 26BB, **Hero calls 26BB**.

River (89.5BB, 2 players) 2♦

BB bets 55.5BB (all in), **Hero?**

We'll begin with Hero's calling range pre-flop. He'll be looking to defend something like the

following to ensure he isn't overfolding or underfolding his opening range vs. the 3-bet.

AA	AKs	AQs	AJs	ATs	A9s	A8s	A7s	A6s	A5s	A4s	A3s	A2s
AKo	KK	KQs	KJs	KTs	K9s	K8s	K7s	K6s	K5s	K4s	K3s	K2s
AQo	KQo	QQ	QJs	QTs	Q9s	Q8s	Q7s	Q6s	Q5s	Q4s	Q3s	Q2s
AJo	KJo	QJo	JJ	JTs	J9s	J8s	J7s	J6s	J5s	J4s	J3s	J2s
ATo	KTo	QTo	JTo	TT	T9s	T8s	T7s	T6s	T5s	T4s	T3s	T2s
A9o	K9o	Q9o	J9o	T9o	99	98s	97s	96s	95s	94s	93s	92s
A8o	K8o	Q8o	J8o	T8o	98o	88	87s	86s	85s	84s	83s	82s
A7o	K7o	Q7o	J7o	T7o	97o	87o	77	76s	75s	74s	73s	72s
A6o	K6o	Q6o	J6o	T6o	96o	86o	76o	66	65s	64s	63s	62s
A5o	K5o	Q5o	J5o	T5o	95o	85o	75o	65o	55	54s	53s	52s
A4o	K4o	Q4o	J4o	T4o	94o	84o	74o	64o	54o	44	43s	42s
A3o	K3o	Q3o	J3o	T3o	93o	83o	73o	63o	53o	43o	33	32s
A2o	K2o	Q2o	J2o	T2o	92o	82o	72o	62o	52o	42o	32o	22

Call

Figure 79 - A Default Defending Range CO vs. BB

This range equates to defending just under half of a fairly wide CO opening range of around 30%. It may be necessary to open the CO tighter depending on the player pool, but this range is definitely fine for more passive micro-stake games. The wider Hero opens, the wider his default defending range vs. the 3-bet.

On the flop we're looking at calling any **Ax** and our better pocket pairs as well as a few good broadway combos with backdoor flush draws. This gives us some balance and prevents Villain from being able to c-bet and then give up on the turn with impunity. If you're keen on practicing this exact process of combo counting, then go ahead and tally up some combos and see how wide we are defending the flop.

On the turn, enough of our flop calling range was **Ax** that we can now fold the rest of our range at this point without folding too much or too little to be balanced on this street. As an aside, most populations tend to underbluff the turn so being a bit tight here is not going to be the end of the world.

So let's focus our attention on the river: the balance decision we're primarily concerned with.

The balancing principle we met earlier can now be extended to cover the situation where Hero is facing a bet.

Balancing Principle 2

Villain is indifferent to bluffing whenever Villain's bet size generates an RFE of X.

AND

Hero is folding X% of his range and calling only hands that beat all of Villain's bluffing range.

In this hand Hero needs to calculate Villain's RFE in the usual fashion. He then proceeds by counting up all of his combos and calling exactly 100 minus Villain's RFE% of those combos deciding which ones are best to call.

So Villain's RFE is (R / R + PG) = 55 / (55 + 89.5) = **38%**

Hero's range contains the suited aces: **AQs, AJs, ATs, A9s, A5s, A4s, A3s** and **A2s** and all of these are 24 combos combined (8 x 3)

Hero's range contains the offsuit aces: **AQo** and **AJo** and these are 18 total (9 x 2).

Hero's range contains the quads **88** (1 combo).

Hero's total river range contains **43 combos**. He needs to fold 38% of these and so needs to call 100 - 38 = 62% of those to make Villain indifferent to bluffing him.

Hero needs to call 43 x 0.62 = **27 combos** (rounded).

Now to decide which ones are best to call. Well, here blockers aren't really an issue. There are no busted draws to consider and so while **ATs** and **A4s** might be as good as each other to call with, we may as well just revert to preferring the one with the most SDV. This spot is simple. Hero will just call the top 27 combos SDV wise.

These are **AQ** (12), **AJ** (12), and then **ATs** (3). That's the balanced strategy solved for. **AT** and above is a call and everything below is a fold. Hero is unexploitable. Bluffing for Villain is 0EV and so doing it too much or too little makes no difference to either player in this one hand (though it may hurt Villain long-term if Hero notices.)

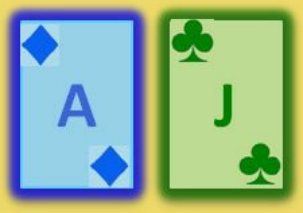
Final Disclaimer: most lower stakes populations, in the experience of myself and my students, do not triple barrel these high dry boards anywhere near enough to be balanced and so the best strategy in reality is likely to overfold our range. Maybe we can fold everything but **AQ** and that strategy will turn out to be higher long-term EV than the balanced one we just created.

So often, our job as poker players is to maximise EV by finding the right spots to deviate from balance and to what extent.

Let's consider a few of examples to round off the chapter. Compare the following two situations:

Hand 135

UTG – Unknown
HJ – Unknown
CO – Unknown
BU – Unknown
SB – Hero
BB – Unknown (0 hands)

Dealt to Hero: 

UTG folds, HJ folds, CO folds, BU folds, **Hero raises to 3BB**, **BB raises to 8.5BB**, **Hero calls 5.5BB**.

Flop (17BB, 2 players)

Hero checks, **BB bets 9.5BB**, **Hero calls 9.5BB**.

Turn (36BB, 2 players)

Hero checks, **BB bets 23BB**, **Hero calls 23BB**.

River (82BB, 2 players)

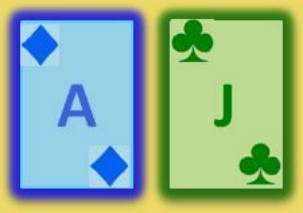
Hero checks, **BB bets 59BB (all in)**, **Hero?**

Have a look at this hand and have a think about what range Hero gets to the river with, then consider

the difference between this and Hand 136 below.

Hand 136

UTG – Unknown
HJ – Unknown
CO – Unknown
BU – Unknown
SB – Hero
BB – Unknown (0 hands)

Dealt to Hero: 

UTG folds, HJ folds, CO folds, BU folds, **Hero raises to 3BB**, **BB raises to 8.5BB**, **Hero calls 5.5BB**.

Flop (17BB, 2 players)

Hero checks, **BB bets 9.5BB**, **Hero calls 9.5BB**.

Turn (36BB, 2 players)

Hero checks, **BB bets 23BB**, **Hero calls 23BB**.

River (82BB, 2 players)

Hero checks, **BB bets 59BB (all in)**, **Hero?**

The difference between these two situations is that in Hand 135, Hero has called the flop and turn with what is pretty far from the top of his range. His hand is low enough down on the river that he should have a comfortable fold to the river shove. Villain's RFE on this shove comes out at 59 / (59 + 82) = **42%**. Hero should call the most suitable **58%** of his range. He has plenty of combos of **QT**, **QJ**, **KQ** and **AQ** that have greater SDV and these hands also do a better job of blocking Villain's value range. **AJ** is not going to qualify as a call. Hero can work this out roughly in-game and come to this conclusion without calculating the exact percentages.

In Hand 136, Hero is very near the top of his range on the flop and the turn and much closer to the top of his range on the river than he was in the hand before. The combos of **QT**, **KQ** and **AQ** no longer exist in his range by the river as Hero has no incentive to get to the river with these hands. This means that **AJ** is very likely to be in the top 58% of his hands suitability wise since he has few combos of better and the **A** does a nice job of blocking **AQ** and **AA** - two of Villain's likely value combos.

Hero folds in Hand 135.

Hero calls in Hand 136.

In the final hand of the chapter below, there are two questions to consider:

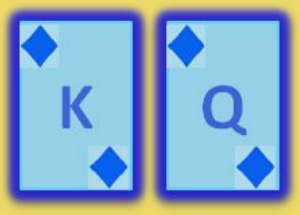
Q1. Is Hero's hand suitable enough to warrant a call from a balanced perspective?

Q2. Does Hero have a reason to adopt some exploitative strategy instead and does this change his decision?

The HUD reverts back to its usual form for when we are not the pre-flop aggressor. The red number is **3Bet Preflop** and the following three brown numbers are **CBet Flop**, **CBet Turn** and **CBet River** respectively.

Hand 137

UTG – Hero
HJ – Unknown
CO – Unknown
BU – Crazy Reg (34/28/**16**/**89**/**67**/**75**) (575 hands)
SB – Unknown
BB – Unknown

Dealt to Hero: 

Hero raises to 3BB, HJ folds, CO folds, **BU raises to 8BB**, SB folds, BB folds, **Hero calls 5BB**.

Flop (17.5BB, 2 players)

Hero checks, **BU bets 12BB**, **Hero calls 12BB**.

Turn (41.5BB, 2 players)

Hero checks, **BU bets 25BB**, **Hero calls 25BB**.

River (91.5BB, 2 players)

Hero checks, **BU bets 55BB (all in)**, **Hero?**

Q1. Villain's RFE on a river bluff is 55 / (55 + 91.5) = **38%**. Hero needs to call 62% of his river

range to play defensively. This hand is certainly fairly low down in Hero's river range and will not creep into the top 62%. It has no club blocker and when quite a few draws get there, there are few worse SDV hands in Hero's range. He has a few combos of flushes, some quads and boats, **AQ** and some but not too much **Jx**. His overall range is strong enough that **KQ** with no club is an easy fold from a balanced perspective.

Q2. But, Hero has a great reason to disregard balance and play his range by dramatically overcalling vs. this Villain. BU is extremely aggressive, likely to have a very wide 3-bet range pre-flop, and likely to be overbluffing every street substantially based on these stats. Hero only seeks the balanced answer when he lacks the information to accurately determine whether he is likely to meet his RE equity target on a call. RE only makes sense to consider when we have some idea whether or not we can muster the sufficient equity to satisfy it.

RE = ATC / (ATC + TP)

RE = 55 /(55 +146.5)

RE = 27%

It seems clear that even on one of the worst rivers possible, Villain's range is likely to contain more bluffs than this measly target requires it to. Hero needs to play exploitatively and call down, even with a hand this low in his range.

Hero calls 55BB

Let's summarise what we've just learned.

The Offensive/Defensive Rule

When Hero has a good idea whether or not he is likely to meet either RE on a call or RFE on a bluff, he should play his range **offensively/exploitatively** post-flop and look at the maths from his own perspective.

When Hero is unsure whether or not he is likely to meet either RE on call or RFE on a bluff, he should play his range **defensively/balanced** post-flop and look at the maths from Villain's perspective.

This rule embodies perhaps the most crucial idea in succeeding at the modern online game. We'll finish with our final rule for the chapter.

Degrees of Exploitation

The more true the conditions below, the more incentive Hero has to be extremely unbalanced and favour vacuum EV.

A: Villain is very unbalanced.
B: The pot is large.
C: The spot is rare.
D: Villain is unaware.

Because there is less of a drawback in the form of hurting long term EV.

A: As we saw in that last hand, the more unbalanced Villain is, the greater the margin by which we'll exceed or fall short of our RE or RFE target. The greater this margin, the higher the EV of the exploitative strategy and the vacuum EV of adopting it.

B: Simple enough. The bigger the pot, the more money we win by adopting the best line in a vacuum. Also big pots happen much less often than small ones so this feeds into the next reason too.

C: The rarer the spot the less opportunity Villain will have to firstly, build up data on how we're playing that spot and secondly, exploit that data. This means that high EV vacuum plays incur much less of a long-term downside in rare spots.

D: Here we can get away with sustained unbalanced play even in more common spots since Villain is not the type of player who is likely to adjust properly or even know what is going on in the first place.

We now know how to be balanced, but we must ensure that we don't succumb to the very common pitfall of being too balanced in spots where there are better alternatives. The above guide can be used as an aid in determining when to remain in the box of balance and when to smash your way out in order to make a lot more precious EV. Balance is our vessel to success as we move through the stakes and exploitative spots become fewer and further between, but exploitative spots will always exist, lest online poker is officially dead.